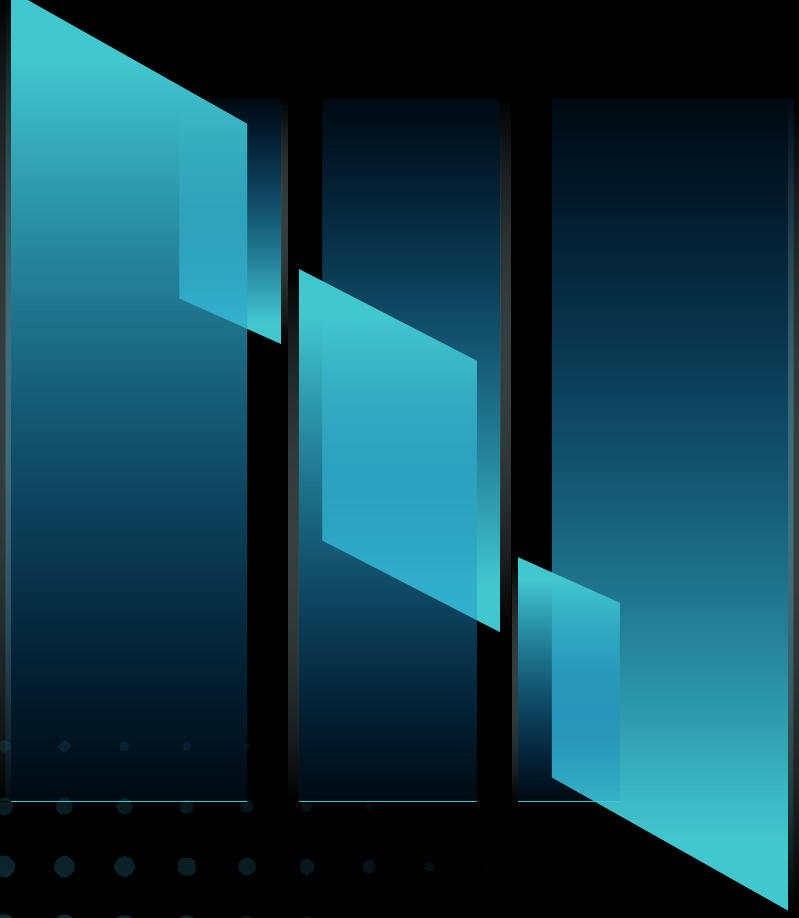


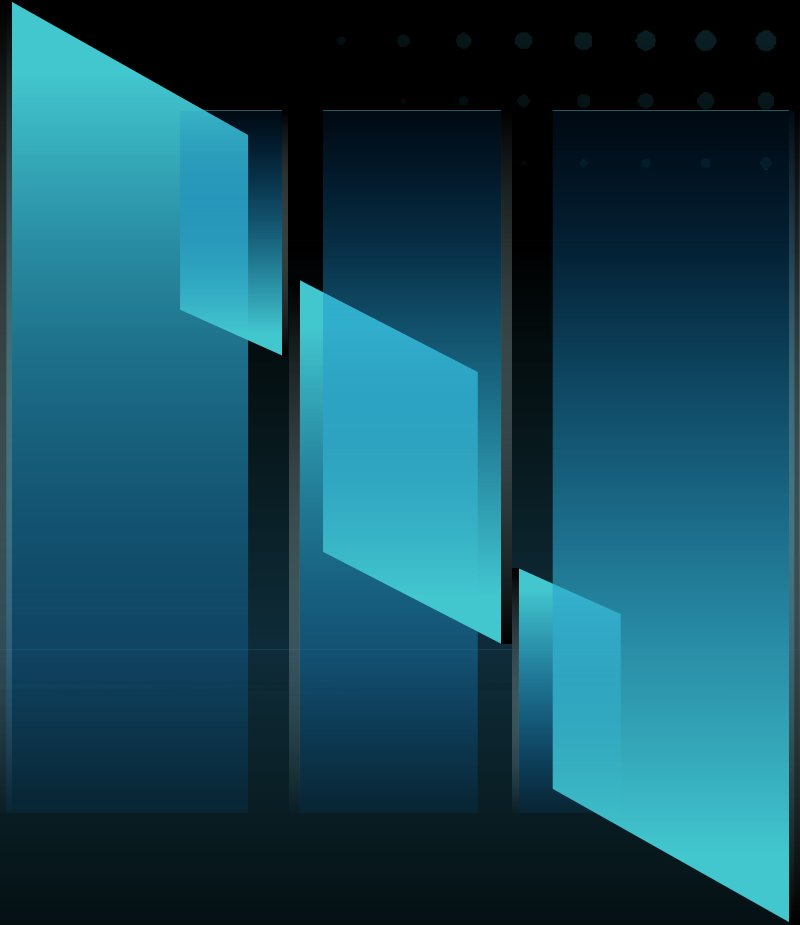
The background features a dark blue gradient. On the left, there are several overlapping, semi-transparent blue geometric shapes, including rectangles and parallelograms, some of which are tilted. On the right side, there is a pattern of small, light blue dots arranged in a grid, which fades out towards the center.

PREDICTIVE MAINTENANCE FOR INDUSTRIAL EQUIPMENT PROJECT

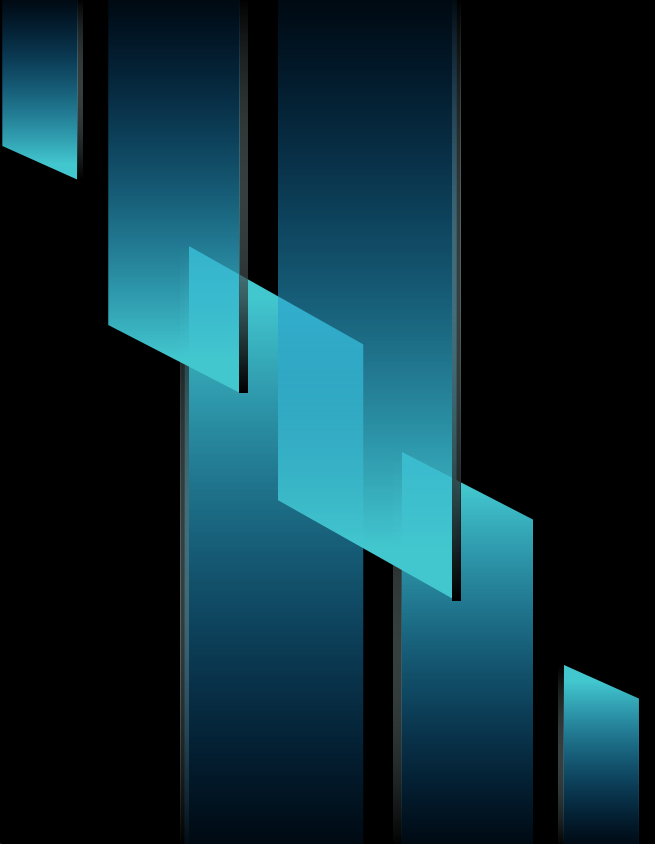
BUILDING A SMARTER, SAFER, AND MORE EFFICIENT MANUFACTURING ECOSYSTEM



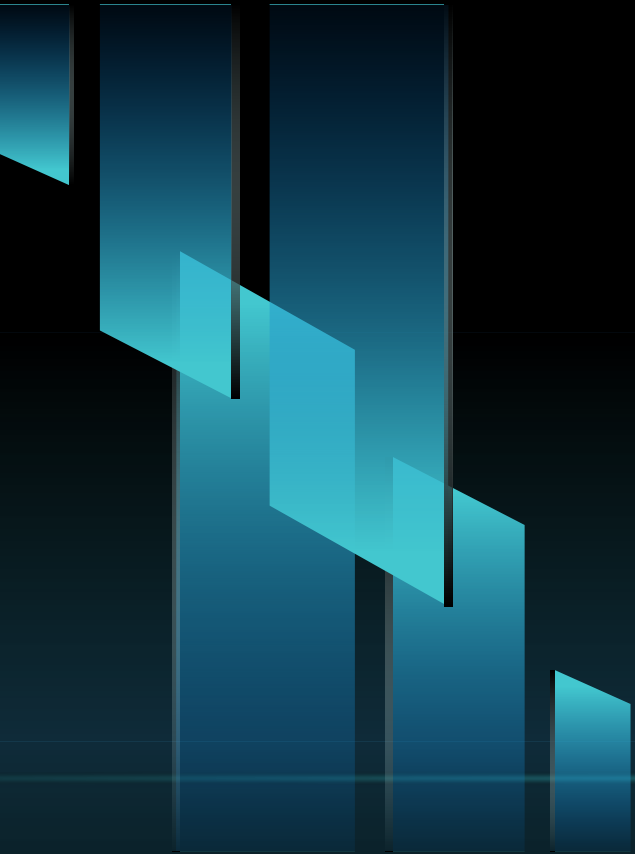
TEAM MEMBERS

- Rawan Essam-ElDin Fahmy
 - Mohamed Sobhy Mohamed
 - Mostafa Kamel Kamel
 - Mohamed Zkaria Abo-Hashim
 - Habiba Ashraf Mohamed
 - Rawan Tarek Hamed
- 

THE BUSINESS CHALLENGE



Industrial operations experience major losses due to:

- Unexpected machine failures that stop production instantly
 - High repair costs due to reactive maintenance
 - Inefficient scheduling, with technicians working under pressure
 - Limited visibility into machine health
 - Data overload with no actionable insights
- 

Unplanned downtime can cost thousands of dollars per minute.

Our goal is to eliminate these surprises.

WHY PREDICTIVE MAINTENANCE?

Traditional maintenance strategies:

- Reactive: fix only after failure
- Preventive: fixed schedule regardless of machine condition

Both approaches are cost-heavy and inefficient.

Predictive maintenance changes everything:

- Uses real-time sensor data
- Predicts failures before they happen
- Extends equipment lifetime
- Reduces unnecessary inspections
- ✓ Less downtime
- ✓ Fewer unexpected failures
- ✓ Better resource planning

OUR DATA FOUNDATION

- We worked with 100,000+ IoT sensor records collected from 50 machines, measuring:
 - Temperature Vibration Pressure
 - Humidity Energy consumption Anomalies
 - Time-based patterns (hour, minute, elapsed time)
- With advanced engineering, we uncovered:
 - Hidden degradation patterns Early signs of stress
 - Correlations between sensors Trends signaling upcoming failures
- This strong dataset made accurate prediction possible.

DATA PREPROCESSING

- Sensor Behavior & Distributions
 - Temperature, vibration, energy roughly normal or right-skewed
 - Outliers preserved (no clipping applied)
 - Most readings within expected operational ranges
- Feature Relationships
 - Moderate correlations between temperature, pressure, energy
 - Vibration largely independent → provides unique predictive info
 - No multicollinearity issues

DATA PREPROCESSING

- Time-Series Trends
 - Gradual drift in vibration and energy for some machines → early maintenance signals
 - Rolling averages highlight long-term trends
- Maintenance Target Analysis
 - Machines needing maintenance show higher averages in: temperature, vibration, energy
 - Boxplots confirm early-warning patterns
 - Failure types (e.g., Vibration Issue, Overheating) align with sensor anomalies

DATA PREPROCESSING

- Advanced Feature Engineering & Analysis
 - Rate-of-change features: temp_diff, vib_diff, pressure_diff → detect sudden spikes
 - Rolling means: temp_roll_mean, vib_roll_mean → smooth noise, highlight drift
 - PCA:
 - PC1: vibration & temperature → thermal-mechanical instability
 - PC2: pressure → load fluctuations
 - PC3: short-term joint disturbances
 - Key predictors: vibration spikes, vibration_diff, temperature drift

THE PREDICTIVE MAINTENANCE PIPELINE

- Our system operates through three integrated machine learning models, each responsible for a critical part of equipment health monitoring:
 1. Maintenance Required? (Binary Classification)
 2. Remaining Useful Life (RUL) (Regression — LSTM Time-Series)
 3. Machine Health Score
- These models work together to provide a full, real-time diagnosis of machine condition.

AIM 1: MAINTENANCE REQUIRED?

What it does?

- Predicts whether a machine is likely to need maintenance soon.

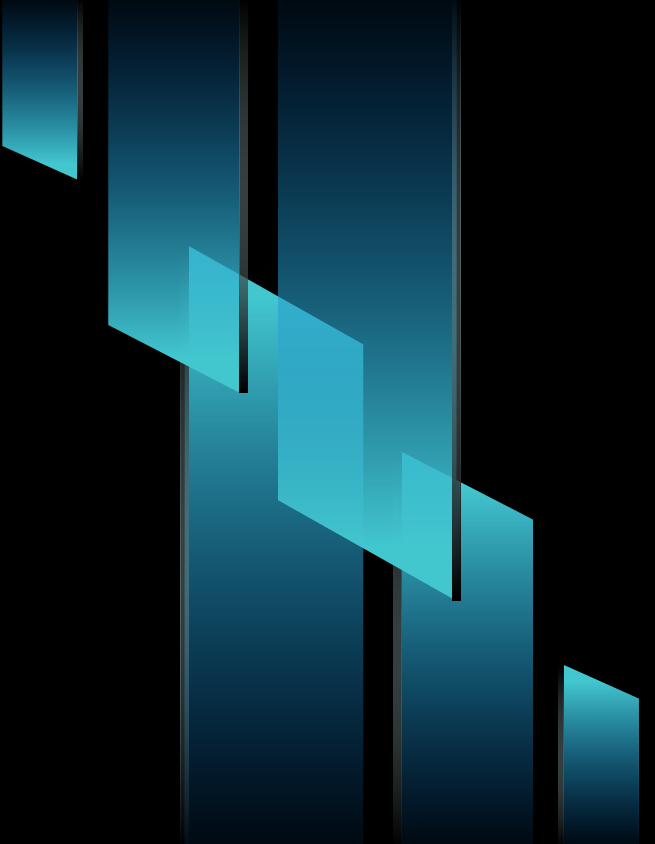
How it helps

- Early alerts prevent sudden stoppages
- Helps managers prioritize high-risk machines
- Optimizes resource allocation

Key Insights

- Machines with high temperature, vibration, and energy consumption had a significantly higher probability of requiring maintenance.

AIM 2: FAILURE TYPE IDENTIFICATION



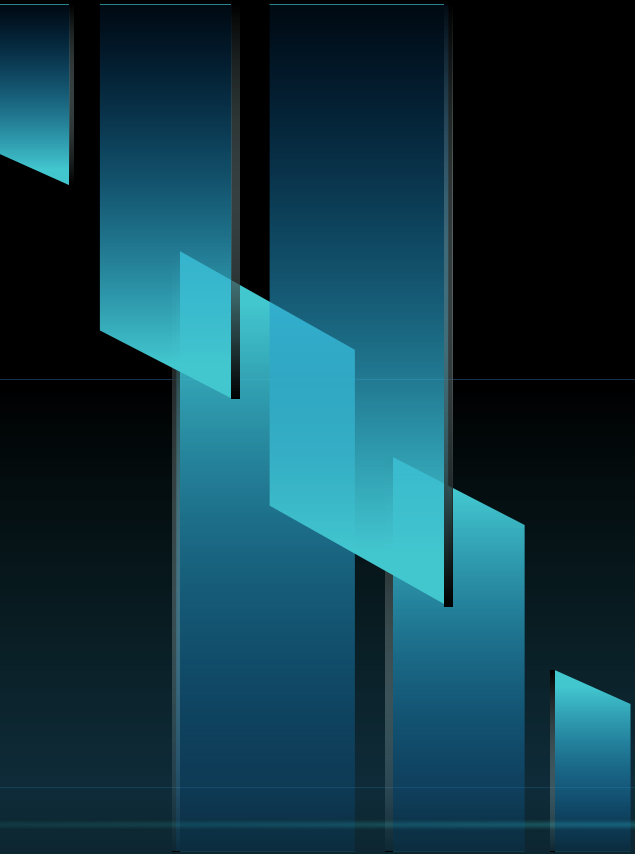
What it does?

- Predicts the cause of the failure (e.g., overheating, vibration issue, pressure imbalance).

How it helps

- Technicians know what to fix before inspection
- Faster troubleshooting
- Reduced machine downtime

Key Insight

- Vibration-related failures were the most common early indicator of stress.
- 

AIM 3: REMAINING USEFUL LIFE (RUL)

PREDICTION

What it does?

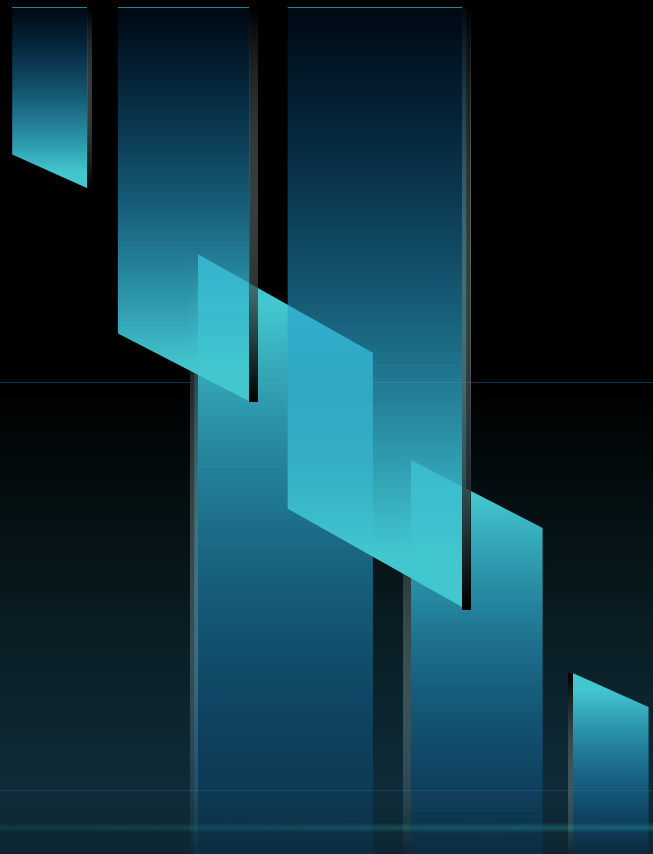
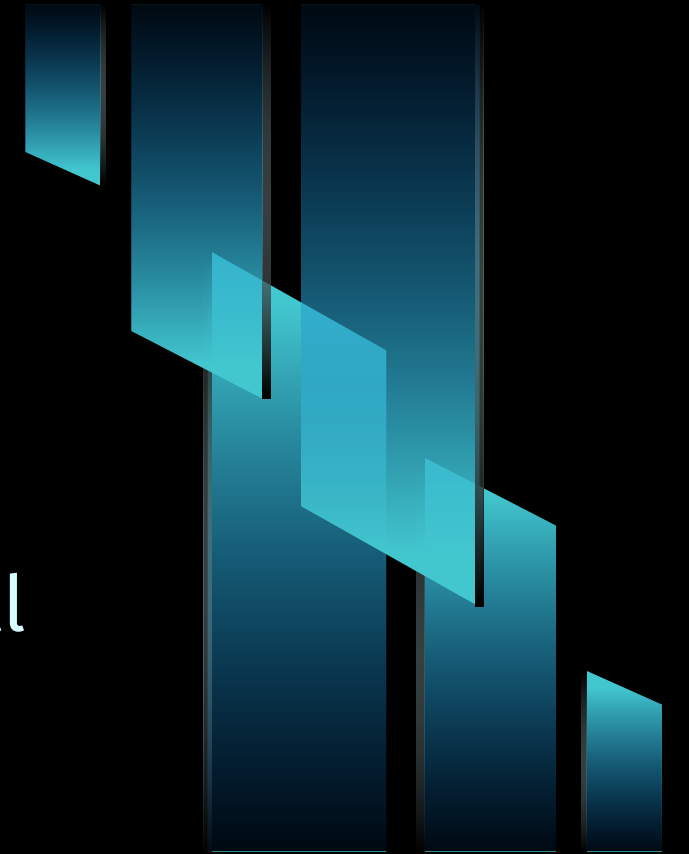
- Uses an LSTM time-series model to forecast how long the machine will run before failure

Why it's crucial

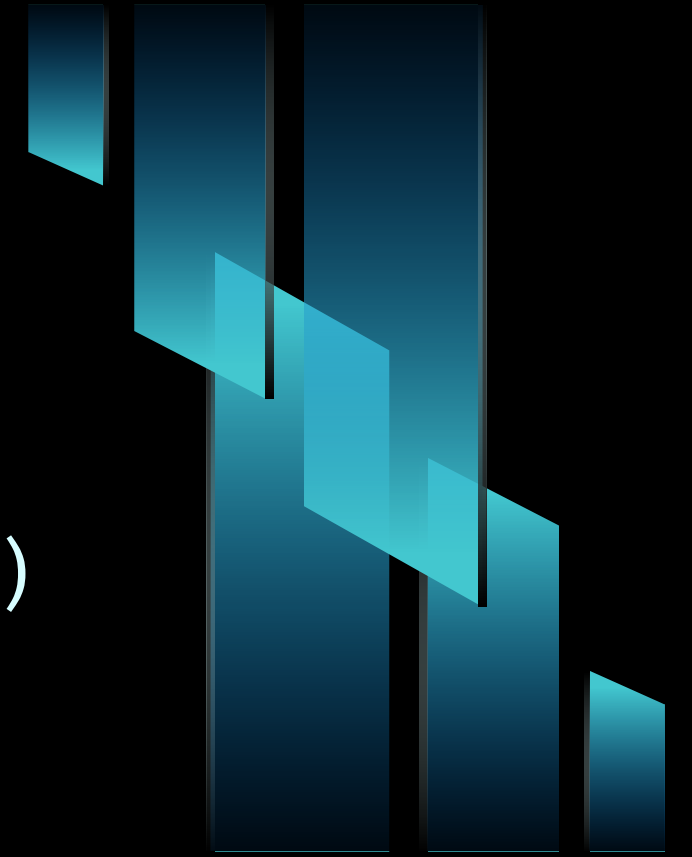
- Enables maintenance scheduling days or weeks in advance
- Reduces emergency repairs
- Supports inventory planning for spare parts

What the model learned

- Vibration spikes shorten RUL
- Temperature drift indicates long-term wear
- Rate-of-change features capture sudden degradation



AIM 4: MACHINE HEALTH SCORE



What it does?

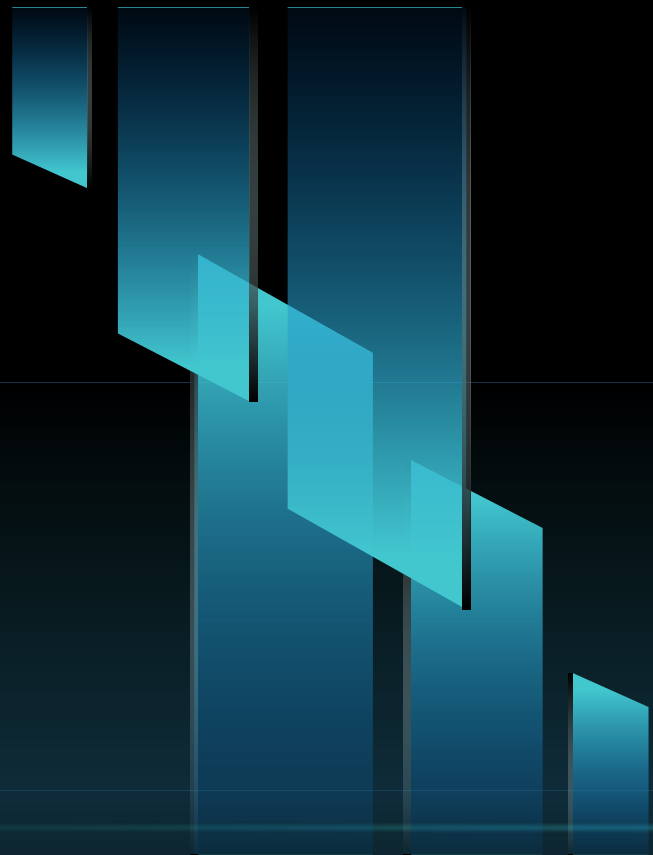
- Uses an LSTM-based model to estimate a real-time health score (0–100) for each machine
- Integrates sensor trends, moving averages, and rate-of-change features

Why it's crucial

- Detects early degradation
- Supports preventive maintenance and alerts

What the model learned

- Spikes in vibration or temperature reduce health
- Persistent deviations indicate gradual wear
- Multi-sensor combination improves reliability

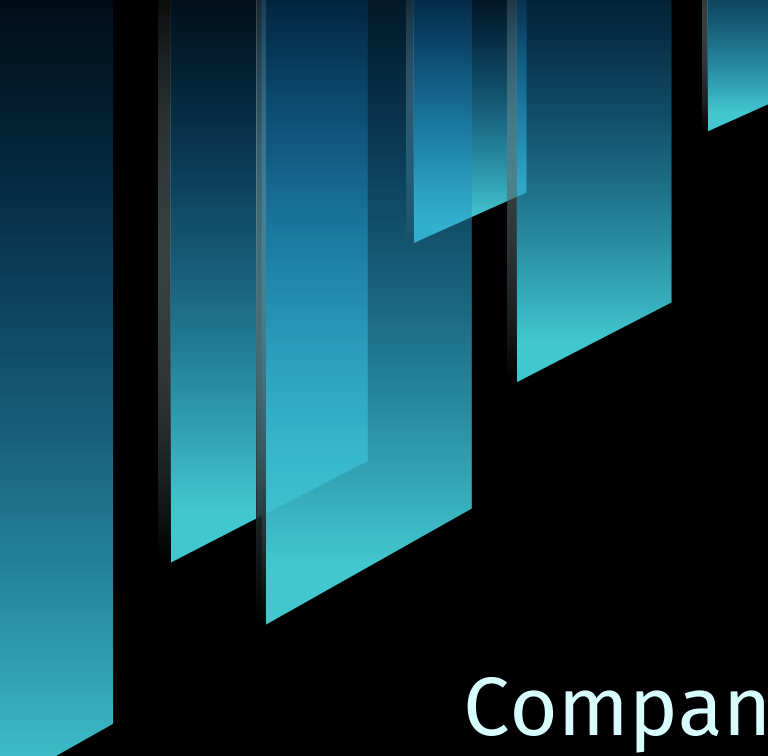


HOW THESE TECHNOLOGIES WORK TOGETHER

Our system provides a complete health assessment:

Technology	Output	Business Value
1. Maintenance Required	“Is the machine at risk?”	Early alert
2. Failure Type	“What is going wrong?”	Faster repairs
3. RUL Prediction	“How long until failure?”	Scheduling & planning
4. Health Score	“How healthy is the machine?”	Preventive maintenance & early warnings

Together, they form a predictive maintenance ecosystem that transforms raw sensor data into actionable insights.

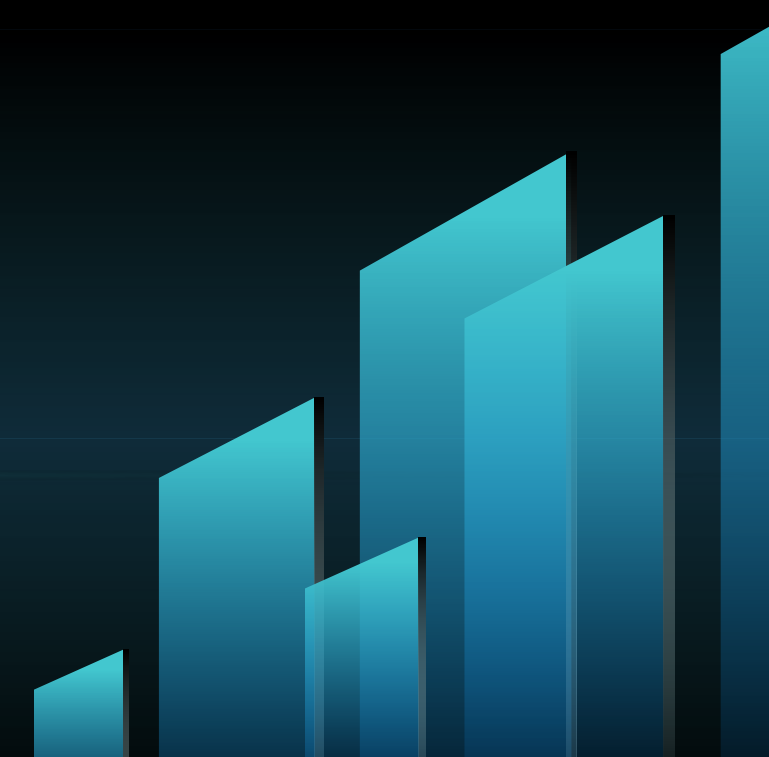


REAL BUSINESS IMPACT

Companies using predictive maintenance see measurable improvements:

- 20–40% reduction in downtime
- 25–30% longer machine lifetime
- Up to 15% savings on maintenance costs
- Higher production output
- Better planning and resource management
- Improved safety and reduced emergency risks

Our models bring the same benefits to this environment.



INDUSTRY APPLICATIONS

This system can be deployed across multiple sectors:

- Manufacturing & assembly lines
- Pumps, motors, rotating equipment
- Oil & gas and mining machinery
- Power generation & utilities
- HVAC and thermal systems
- Logistics & automated warehouses

Any machine with sensors can benefit from our predictive engine.

COMPETITIVE ADVANTAGES

What makes our system stand out:

- Advanced LSTM time-series architecture for accurate RUL prediction
- Strong feature engineering capturing long-term and short-term degradation
- Automated anomaly detection
- Scalable, real-time ready pipelines
- Integration with dashboards, cloud systems, and IoT devices

Our solution is not just accurate — it is practical for industrial deployment.

FINAL SUMMARY

We built a complete predictive maintenance solution that:

- Provides early warnings
- Identifies failure types
- Predicts machine lifetime
- Minimizes downtime
- Cuts costs
- Increases operational efficiency

This system empowers organizations to move from reactive to predictive and ultimately optimized operations.

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THANK YOU